

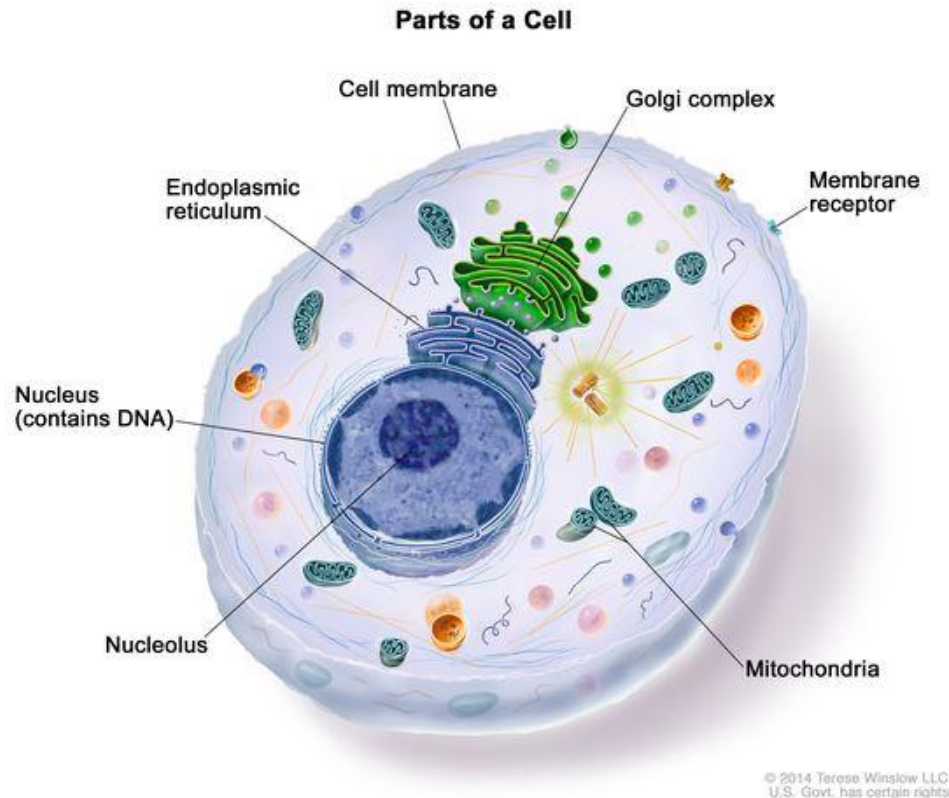
How do cells become
cancerous?

What are we made of?

37 000 000 000 000

37 trillion!

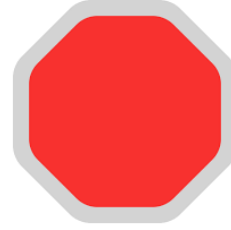
**A human body has 300 times more cells
than there are stars in our entire galaxy!!**



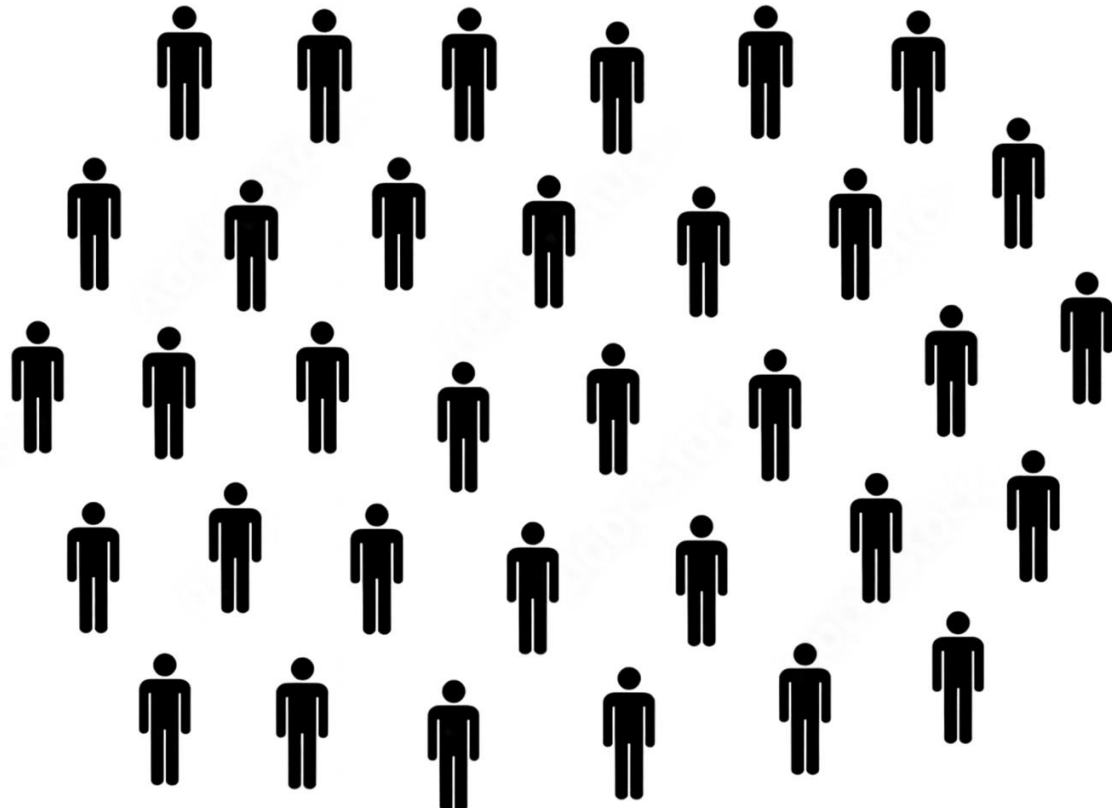
<https://www.cancer.gov/publications/dictionaries/cancer-terms/def/cell>

<https://science.nasa.gov/missions/hubble/supernova-blast-bonanza-in-nearby-galaxy/>

Cancerous tumors get crowded and disorganized



Normal cells

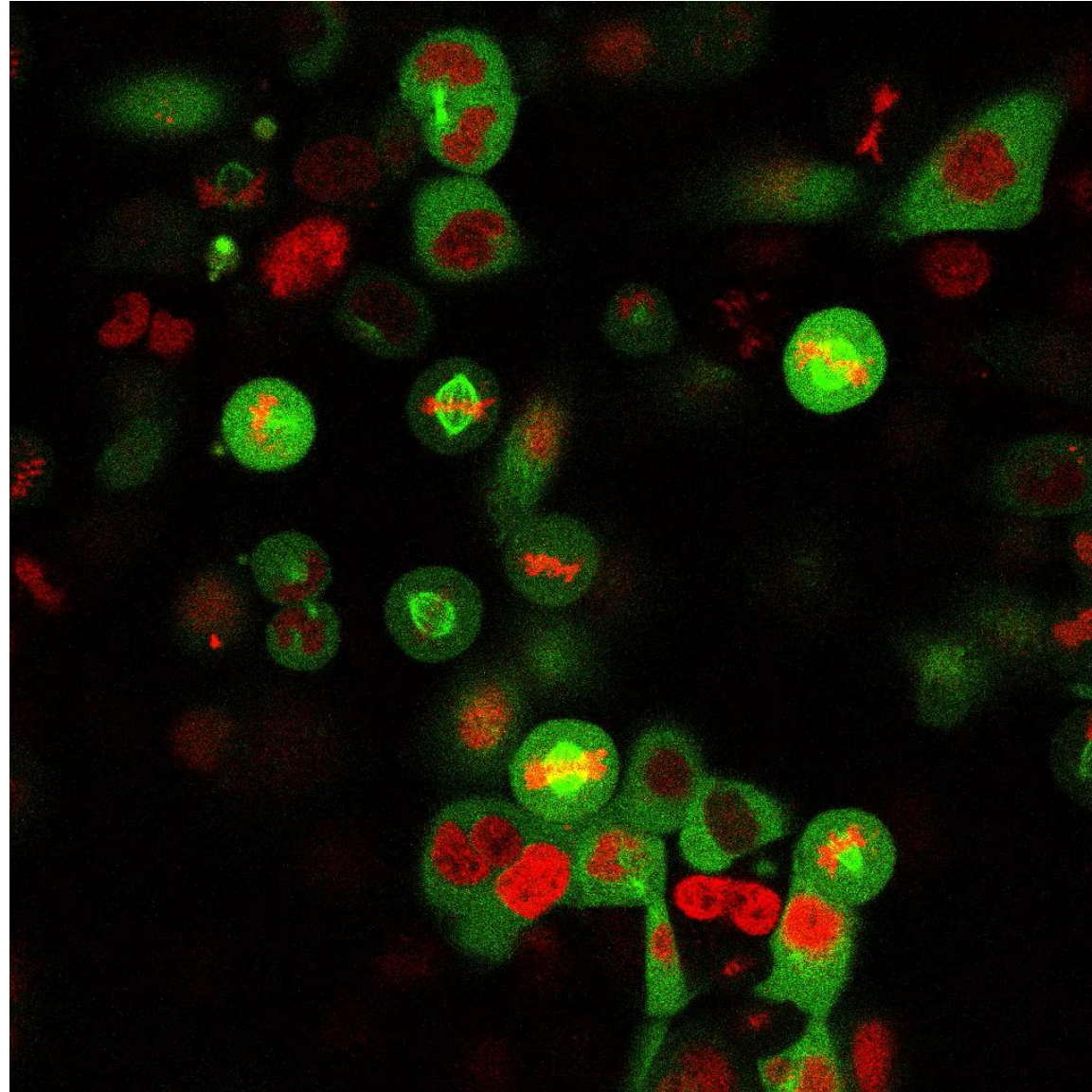


Cancer cells

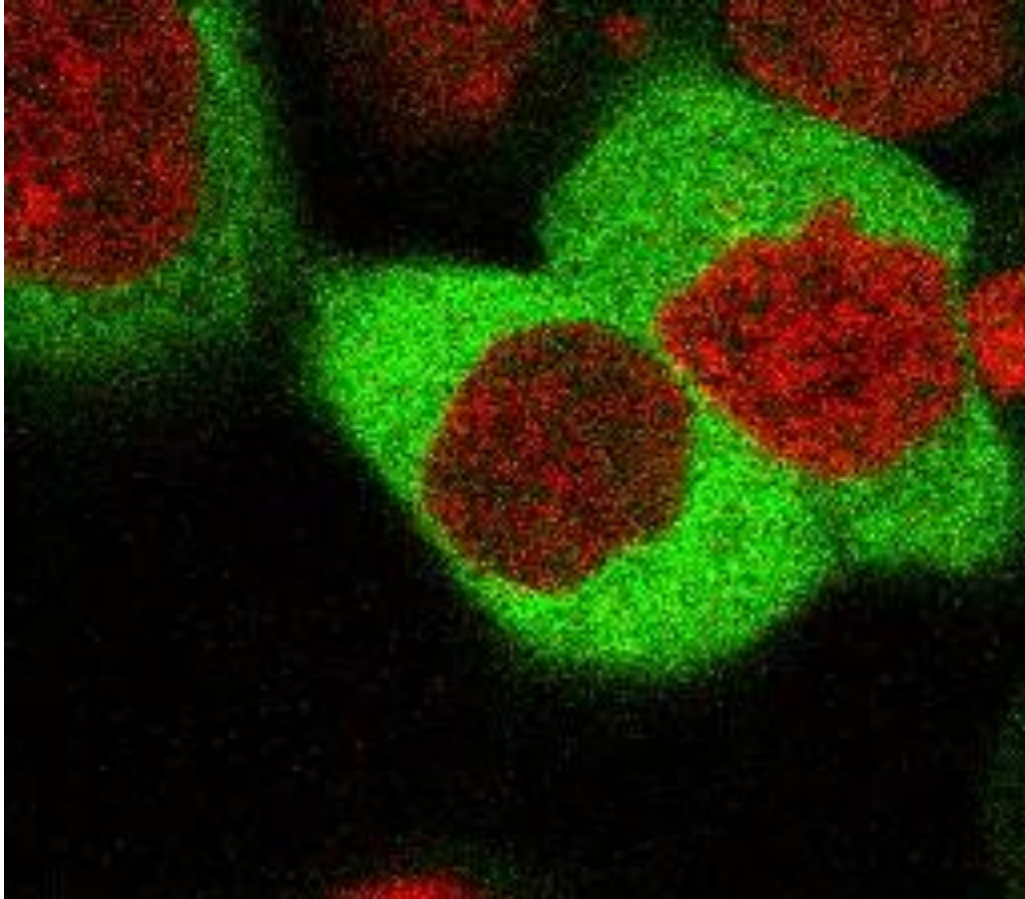


Real term: loss of contact inhibition

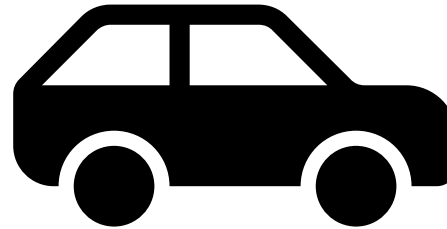
We can color the cells!



Cancer cells reproduce much faster than the body needs



Normal cells



Cancer cells



Real term: sustained proliferative signaling

Cancer cells have a messed-up life cycle

Normal cells



Cancer cells



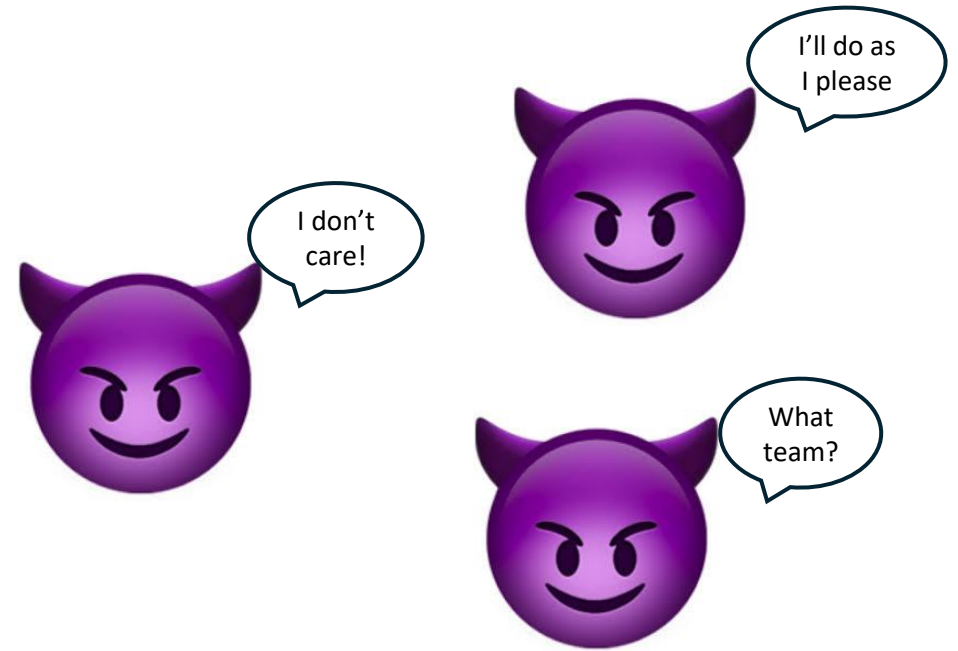
Real term: replicative immortality

They are terrible team players because they become selfish

Normal cells



Cancer cells



Real term: evading growth suppressors

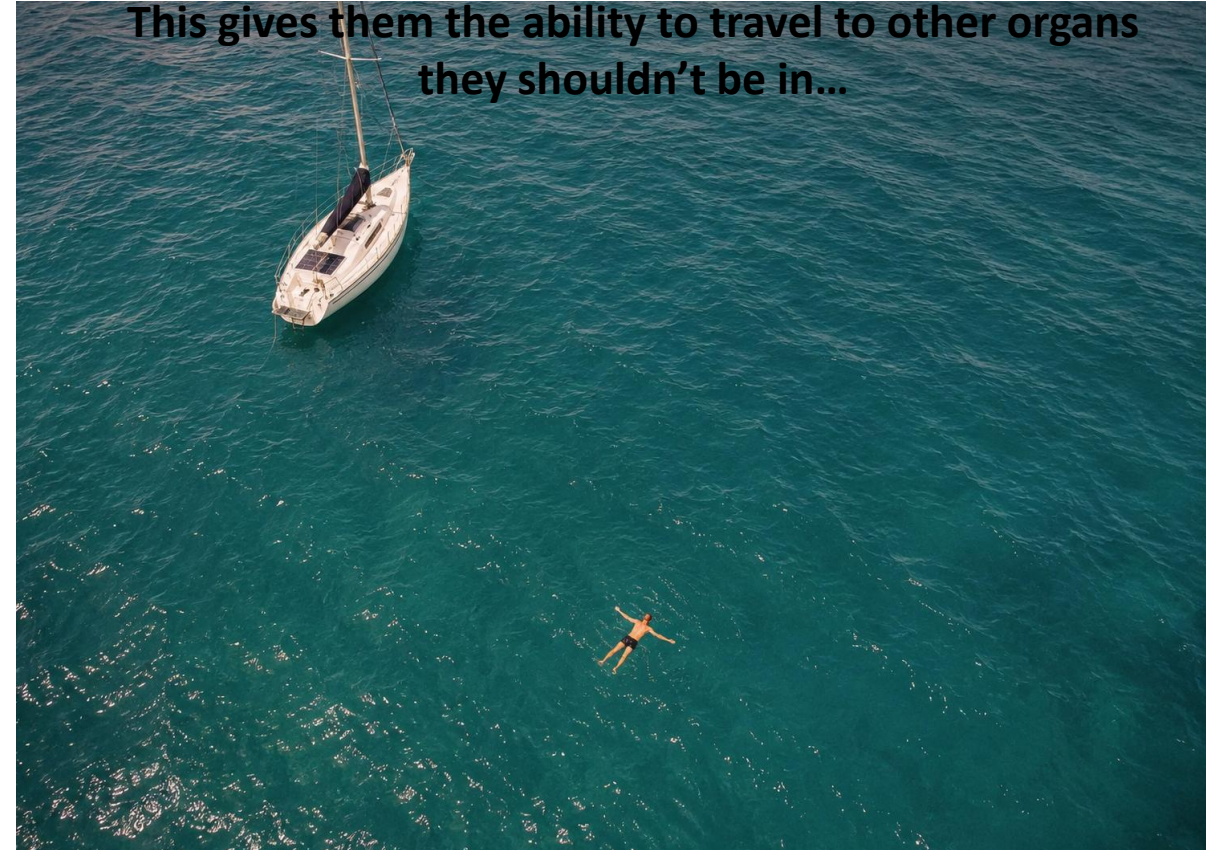
Unlike most normal cells, they can live anchor-free



Normal cells



Cancer cells



Real term: activating invasion & metastasis